



2014 HSC Course Assessment Task 3 (Trial Examination)

Wednesday June 18, 2014

- Working time – 3 hours.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

- Mark your answers on the answer sheet provided (numbered as page 13)

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

Class (please ✓)

○ 12M4C – Mr Ireland

[illegible]

Section I

10 marks

Attempt Question 1 to 10

Allow approximately 15 minutes for this section

Mark your answers on the answer sheet provided.

Questions

Marks

1. Which of the following represents the region defined by the upper part of the semicircle in the Argand diagram with centre $(1, 1)$ and radius 1, cut off by the line $y = x$? 1

(A) $|z + 1 + i| \leq \sqrt{2}$ and $\arg(z) \leq \frac{\pi}{4}$

(B) $|z - 1 - i| \leq 1$ and $\arg(z) \geq \frac{\pi}{4}$

(C) $|z + 1 + i| \leq 1$ and $\arg(z) \geq \frac{\pi}{4}$

(D) $|z - 1 - i| \leq \sqrt{2}$ and $\arg(z) \leq \frac{\pi}{4}$

2. Which of the following shapes represents the locus of the point P representing the complex number z , moving in the Argand diagram such that 1

$$|z - 4i| = \operatorname{Arg}(\sqrt{3} + i) + |z + 4i|$$

(A) Parabola

(B) Ellipse

(C) Hyperbola

(D) Circle

3. Let $z = \sqrt{48} - 4i$. What is the value of $\operatorname{Arg}(z^7)$? 1

(A) $-\frac{2\pi}{3}$

(B) $\frac{2\pi}{3}$

(C) $\frac{5\pi}{6}$

(D) $\frac{-5\pi}{6}$

4. What is the eccentricity of the hyperbola $3x^2 - 4y^2 = 1$? 1

(A) $\frac{\sqrt{7}}{2}$

(B) $\frac{\sqrt{7}}{\sqrt{3}}$

(C) $\frac{1}{2}$

(D) $\frac{1}{\sqrt{3}}$

5. Which of the following integrals does NOT involve integration by parts? 1

(A) $\int 7xe^{-x^2} dx$

(B) $\int \frac{\ln x}{x^2} dx$

(C) $\int x^2 \sin x dx$

(D) $\int e^x \cos x dx$

6. Which of the following expressions will lead to the location of the vertical tangent(s) to the graph of $x^4 + y^4 = 4xy$? 1

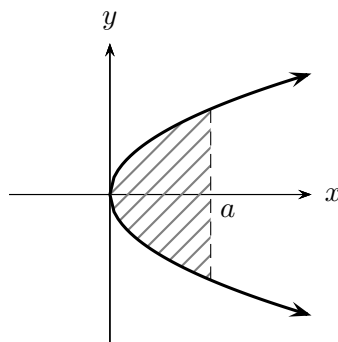
(A) $x^3 + y = 0$

(B) $x^3 - y = 0$

(C) $x - y^3 = 0$

(D) $x + y^3 = 0$

7. A solid is formed by rotating the region enclosed by the parabola $y^2 = 4ax$, its vertex $(0,0)$ and the line $x = a$ about the y axis. 1



Which of the following integrals gives the volume of this area by *slicing*?

- (A) $2\pi\sqrt{a} \int_0^a z^{\frac{3}{2}} dz$
- (B) $4\pi\sqrt{a} \int_0^a z^{\frac{3}{2}} dz$
- (C) $\pi \int_0^{2a} \left(a^2 - \frac{z^4}{16a^2} \right) dz$
- (D) $2\pi \int_0^{2a} \left(a^2 - \frac{z^4}{16a^2} \right) dz$

8. Without evaluating the integrals, which of the following is greater than zero? 1

- (A) $\int_{-1}^1 \tan^{-1}(\sin x) dx$
- (B) $\int_{-1}^1 \frac{2x}{\sin^2 x} dx$
- (C) $\int_{-1}^1 \left((e^x)^3 + x^7 \right) dx$
- (D) $\int_{-1}^1 \frac{x^5}{\cos^3 x} dx$

9. An ellipse has foci $(0, -3)$ and $(0, 5)$.

1

Which of the following could be the equation of the ellipse?

(A) $\frac{x^2}{8} + \frac{(y-1)^2}{12} = 1$

(B) $\frac{x^2}{12} + \frac{(y-1)^2}{8} = 1$

(C) $\frac{x^2}{9} + \frac{(y-1)^2}{25} = 1$

(D) $\frac{x^2}{8} + \frac{(y+1)^2}{12} = 1$

10. The polynomial equation $P(x) = 0$ has real coefficients and has roots that include $x = 4i - 3$, $x = -3$ and $x = -4i + 3$. What is the smallest possible degree of $P(x)$?

1

(A) 2

(B) 3

(C) 4

(D) 5

Examination continues overleaf...

Section II

90 marks

Attempt Questions 11 to 16

Allow approximately 2 hours and 45 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Glossary

- $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3\}$ – set of all integers.
- $\mathbb{Z}^+$ – all positive integers (excludes zero)
- $\mathbb{R}$ – set of all real numbers

Question 11 (15 Marks)

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Marks

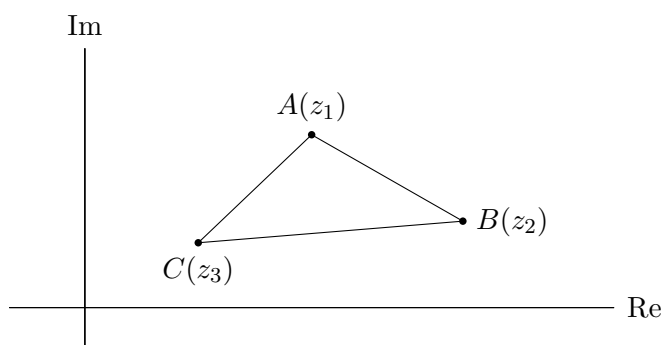
- (a) i. Find the partial fraction decomposition of $\frac{1}{x^2 - 1}$. 2
- ii. Hence or otherwise, evaluate $\int \frac{x^2 + 1}{x^2 - 1} dx$. 3
- (b) i. Given $I_n = \int_0^1 x^n 2^x dx$ ($n \in \mathbb{Z}^+$), show that 3
- $$I_n = \frac{2}{\ln 2} - \frac{n}{\ln 2} I_{n-1}$$
- ii. Hence evaluate $\int_0^1 x^3 2^x dx$. 3
- (c) Evaluate $\int \frac{x}{x^2 + 2x + 10} dx$, giving your answer in simplest form. 4

Question 12 (15 Marks)

Commence a NEW page.

Marks

- (a)
- i. Find the three roots of $z^3 - 1 = 0$ in modulus-argument form. **3**
 - ii. Write each of the complex roots in the form $x + iy$. **2**
 - iii. If one of the complex roots is ω , find the area of the triangle formed by 1, ω and ω^2 . **1**
 - iv. Show that $1 + \omega + \omega^2 = 0$. **2**
 - v. Evaluate $(1 + \omega)(1 + \omega^2)(1 + \omega^5)(1 + \omega^8)(1 + \omega^{11})$ **3**
- (b) A , B and C are the points that represent the complex number numbers z_1 , z_2 and z_3 on the Argand diagram. **4**



Prove that if

$$\frac{z_2 - z_3}{z_1 - z_3} = \frac{z_1 - z_3}{z_1 - z_2}$$

then $\triangle ABC$ is equilateral.

Hint: To commence, extend lengths AC , AB and BC to real axis, and use angles.

Question 13 (15 Marks)

Commence a NEW page.

Marks

- (a) $P(a \cos \theta, b \sin \theta)$ is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

i. Show that the equation of the normal to P is:

3

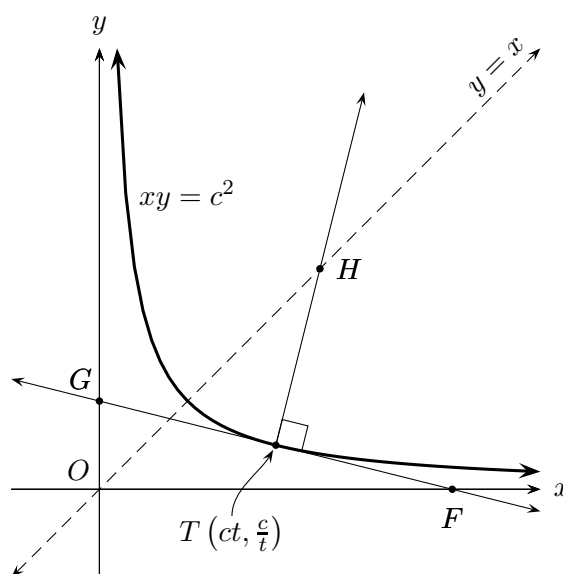
$$ax \sin \theta - by \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$$

- ii. G is the point where this normal meets the x axis. N is the foot of the perpendicular from P to the x axis, O is the origin and e is the eccentricity.

2

Show that $\frac{OG}{ON} = e^2$.

- (b) The tangent to the hyperbola $xy = c^2$ at the point $T\left(ct, \frac{c}{t}\right)$ meets the x and y axes at F and G respectively, and the normal at T meets the line $y = x$ at H .



- i. Show that the tangent at T is

3

$$x + t^2y = 2ct$$

- ii. Show that the normal at T is

2

$$t^3x - ty = c(t^4 - 1)$$

- iii. Prove that $FH \perp HG$.

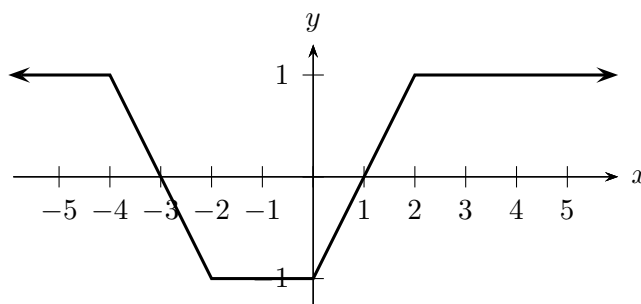
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Question 14 (15 Marks)

Commence a NEW page.

Marks

- (a) The following diagram shows the sketch of the function $y = f(x)$.



On separate diagrams of $\frac{1}{3}$ page each, carefully sketch:

- i. $y = f(|x|)$. **2**
 - ii. $y = \frac{1}{f(x)}$. **2**
 - iii. $y = \cos^{-1}(f(x))$. **2**
- (b) Given $x \in \mathbb{R}$, and $\lceil x \rceil$ be a real number that is the smallest integer that is greater than, or equal to x .
- i. Evaluate $\lceil 2.5 \rceil$ and $(2.5 + \lceil 2.5 \rceil)^2$. **2**
 - ii. Sketch a graph of $y = \lceil x \rceil + (x + \lceil x \rceil)^2$. **3**
- (c) By using the substitution $x = 2 \tan \theta$, evaluate the definite integral **4**

$$\int \frac{dx}{(4 + x^2)^{\frac{3}{2}}}$$

Question 15 (15 Marks)

Commence a NEW page.

Marks

- (a) A solid is formed by rotating the region bounded $y = \sqrt{x}$, the x axis and the line $x = 4$, about the line $x = 4$.

- i. By drawing a diagram and taking slices perpendicular to the axis of rotation, show that the element of volume δV is **2**

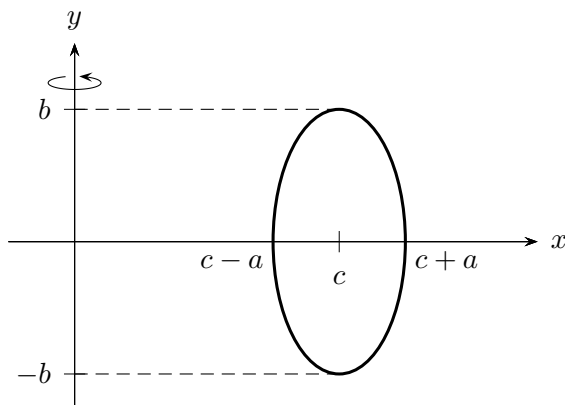
$$\delta V = \pi (16 - 8y^2 + y^4) \delta y$$

- ii. Hence or otherwise, find the volume generated. **2**

- (b) i. Sketch the curve $x = \sqrt{b^2 - y^2}$, and hence explain why **2**

$$\int_0^b \sqrt{b^2 - y^2} dy = \frac{\pi b^2}{4}$$

- ii. The ellipse $\frac{(x-c)^2}{a^2} + \frac{y^2}{b^2} = 1$, where $b > a$ and $c > a$ is shown in the diagram. The region bounded by the ellipse is rotated about the y axis to form a ring. **4**



By taking slices perpendicular to the y axis, show that the ring has volume $2abc\pi^2$.

- (c) i. Show that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$. **2**

- ii. Hence or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \frac{e^{\sin x}}{e^{\sin x} + e^{\cos x}} dx$. **3**

Question 16 (15 Marks)

Commence a NEW page.

Marks

- (a) The equation $x^3 - 4x^2 + 5x + 2 = 0$ has roots α , β and γ .

i. Show that $\alpha^2 + \beta^2 + \gamma^2 = 6$.

1

ii. Find the value of $\alpha^3 + \beta^3 + \gamma^3$.

2

- (b) A polynomial $P(x)$ is divided by $x^2 - a^2$ (where $a \neq 0$) and the remainder is $px + q$.

- i. Show that

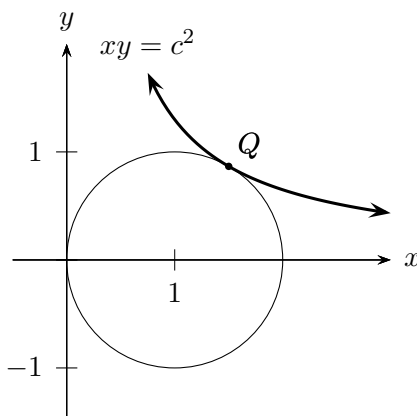
3

$$p = \frac{1}{2a}[P(a) - P(-a)] \quad \text{and} \quad q = \frac{1}{2}[P(a) + P(-a)]$$

- ii. Find the remainder when $P(x) = x^n - a^n$ for $n \in \mathbb{Z}^+$, is divided by $x^2 - a^2$.

2

- (c) The hyperbola $xy = c^2$ touches the circle $(x - 1)^2 + y^2 = 1$ at the point Q .



- i. By considering the diagram provided or otherwise, deduce that the equation $x^2(x - 1)^2 + c^4 = x^2$ has a repeated real root $\beta > 0$, as well as two non-real complex roots.

2

- ii. Find the values of β and c^2 .

3

Hint: Consider a property of β being a repeated root of $P(x) = 0$.

- iii. Find the equation of the common tangent to the hyperbola and the circle at Q .

2**End of paper.**

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

STUDENT NUMBER:

Class (please ✓)

☐ 12M4A – Ms Ziazaris

☐ 12M4B – Mr Lam

☐ 12M4C – Mr Ireland

- | | | | | | |
|-----------|---|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 2 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 3 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 4 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 5 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 6 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 7 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 8 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 9 | – | ☐ A | ☐ B | ☐ C | ☐ D |
| 10 | – | ☐ A | ☐ B | ☐ C | ☐ D |

Suggested Solutions

Section I

(b) i. (3 marks)

1. (B) 2. (C) 3. (C) 4. (A) 5. (A) 6. (C) 7. (D)

8. (C) 9. (C) 10. (D)

Question 11 (Lam)

(a) i. (2 marks)

$$\frac{1}{(x-1)(x+1)} \equiv \frac{A}{x-1} + \frac{B}{x+1}$$

$$1 \equiv A(x+1) + B(x-1)$$

- When $x = 1$,

$$1 = A(1+1) + B(1-1)$$

$$\therefore A = \frac{1}{2}$$

- When $x = -1$,

$$1 = A(-1+1) + B(-1-1)$$

$$\therefore B = -\frac{1}{2}$$

$$\therefore \frac{1}{x^2-1} = \frac{\frac{1}{2}}{x-1} - \frac{\frac{1}{2}}{x+1}$$

ii. (3 marks)

$$\int \frac{x^2+1}{x^2-1} dx = \int \frac{x^2-1+2}{x^2-1} dx$$

$$= \int \left(1 + \frac{2}{x^2-1} \right) dx$$

$$= \int \left(1 + 2 \left(\frac{\frac{1}{2}}{x-1} - \frac{\frac{1}{2}}{x+1} \right) \right) dx$$

$$= \int \left(1 + \frac{1}{x-1} - \frac{1}{x+1} \right) dx$$

$$= x + \ln(x-1) - \ln(x+1) + C$$

$$= x + \ln \left(\frac{x-1}{x+1} \right) + C$$

$$I_n = \int_0^1 \underbrace{x^n}_{=u} \underbrace{2^x}_{=dv} dx$$

$$u = x^n \quad v = \frac{1}{\ln 2} 2^x$$

$$du = nx^{n-1} \quad dv = 2^x$$

$$\therefore I_n = \left[\frac{x^n 2^x}{\ln 2} \right]_0^1 - \int_0^1 nx^{n-1} \times \frac{1}{\ln 2} 2^x dx$$

$$= \left(\frac{2}{\ln 2} - 0 \right) - \frac{n}{\ln 2} \int_0^1 x^{n-1} 2^x dx$$

$$= \frac{2}{\ln 2} - \frac{n}{\ln 2} I_{n-1}$$

ii. (3 marks)

$$I_0 = \int_0^1 x^0 2^x dx = \int_0^1 2^x dx$$

$$= \frac{1}{\ln 2} \left[2^x \right]_0^1$$

$$= \frac{1}{\ln 2} (2 - 1) = \frac{1}{\ln 2}$$

$$I_1 = \frac{2}{\ln 2} - \frac{1}{\ln 2} I_0$$

$$= \frac{2}{\ln 2} - \frac{1}{\ln 2} \left(\frac{1}{\ln 2} \right)$$

$$= \frac{2}{\ln 2} - \frac{1}{\ln^2 2}$$

$$I_2 = \frac{2}{\ln 2} - \frac{2}{\ln 2} I_1$$

$$= \frac{2}{\ln 2} - \frac{2}{\ln 2} \left(\frac{2}{\ln 2} - \frac{1}{\ln^2 2} \right)$$

$$= \frac{2}{\ln 2} - \frac{4}{\ln^2 2} + \frac{2}{\ln^3 2}$$

$$I_3 = \frac{2}{\ln 2} - \frac{3}{\ln 2} I_2$$

$$= \frac{2}{\ln 2} - \frac{3}{\ln 2} \left(\frac{2}{\ln 2} - \frac{4}{\ln^2 2} + \frac{2}{\ln^3 2} \right)$$

$$= \frac{2}{\ln 2} - \frac{6}{\ln^2 2} + \frac{12}{\ln^3 2} - \frac{6}{\ln^4 2}$$

(c) (4 marks)

$$\begin{aligned}
& \int \frac{x}{x^2 + 2x + 10} dx \\
&= \frac{1}{2} \int \frac{2x + 2 - 2}{x^2 + 2x + 10} dx \\
&= \frac{1}{2} \int \frac{2x + 2}{x^2 + 2x + 10} dx - \int \frac{1}{(x^2 + 2x + 1) + 9} dx \\
&= \frac{1}{2} \ln(x^2 + 2x + 10) - \int \frac{1}{(x + 1)^2 + 9} dx \\
&= \frac{1}{2} \ln(x^2 + 2x + 10) - \frac{1}{3} \tan^{-1} \left(\frac{x + 1}{3} \right) + C
\end{aligned}$$

iv. (2 marks)

$$\begin{aligned}
& \omega^3 - 1 = 0 \\
& (\omega - 1)(\omega^2 + \omega + 1) = 0
\end{aligned}$$

Given $\text{Im}(\omega) \neq 0$,**Question 12** (Ireland)

(a) i. (3 marks)

$$\begin{aligned}
& z^3 - 1 = 0 \\
& z^3 = 1 = \cos(2k\pi) + i \sin(2k\pi)
\end{aligned}$$

$$\therefore \omega^2 + \omega + 1 = 0$$

Applying De Moivre's Theorem,

$$z = \cos\left(\frac{2k\pi}{3}\right) + i \sin\left(\frac{2k\pi}{3}\right)$$

v. (3 marks)

- $k = 0, z = 1.$
- $k = 1, z = \cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right).$
- $k = -1,$
 $z = \cos\left(-\frac{2\pi}{3}\right) + i \sin\left(-\frac{2\pi}{3}\right).$

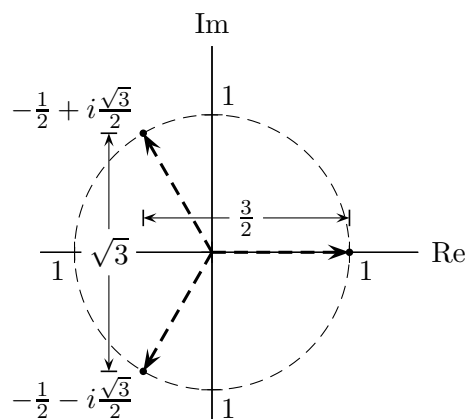
$$\omega^2 + \omega + 1 = 0$$

ii. (2 marks)

$$\therefore \omega^2 = -(\omega + 1)$$

$$-\frac{1}{2} \pm i \frac{\sqrt{3}}{2}$$

iii. (1 mark)

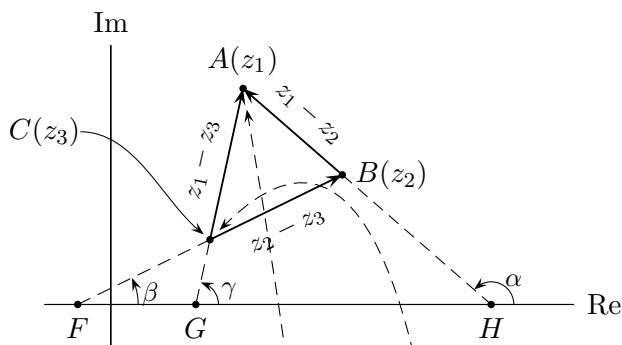


$$A = \frac{1}{2}bh = \frac{1}{2} \times \frac{3}{2} \times \sqrt{3} = \frac{3\sqrt{3}}{4}$$

Now examine expression:

$$\begin{aligned}
& (1 + \omega)(1 + \omega^2)(1 + \omega^5)(1 + \omega^8)(1 + \omega^{11}) \\
&= -\omega^2 (\cancel{1} - (\cancel{1} + \omega)) \left(1 + \cancel{\omega^2}\omega^2\right) \\
&\quad \times \left(1 + \cancel{\omega^4}\omega^2\right) \left(1 + \cancel{\omega^6}\omega^2\right) \\
&= \cancel{\omega^2} (1 + \omega^2)^3 \\
&= (\cancel{1} - (\cancel{1} + \omega))^3 \\
&= -1
\end{aligned}$$

(b) (4 marks)



where

- $\alpha = \arg(z_1 - z_2)$
- $\beta = \arg(z_2 - z_3)$
- $\gamma = \arg(z_1 - z_3)$

Given

$$\frac{z_2 - z_3}{z_1 - z_3} = \frac{z_1 - z_3}{z_1 - z_2} \quad (b)$$

Then by taking arguments,

$$\arg\left(\frac{z_2 - z_3}{z_1 - z_3}\right) = \arg\left(\frac{z_1 - z_3}{z_1 - z_2}\right)$$

$$\begin{aligned} \therefore \arg(z_2 - z_3) - \arg(z_1 - z_3) \\ = \arg(z_1 - z_3) - \arg(z_1 - z_2) \end{aligned}$$

From diagram,

$$\beta - \gamma = \gamma - \alpha \quad (\diamond)$$

- Now in $\triangle FCG$,
 - $\angle ACB = \angle FCG$ (vertically opposite)
 - Hence $\beta + \angle FCG = \gamma$, and

$$\angle FCG = \angle ACB = \gamma - \beta$$

- In $\triangle AGH$
 - $\gamma + \angle CAB = \alpha$ (exterior $\angle$ of $\triangle$)
 - Hence $\angle CAB = \alpha - \gamma$. But from $(\diamond)$,

$$\begin{aligned} \alpha - \gamma &= \gamma - \beta \\ \therefore \angle CAB &= \gamma - \beta = \angle ACB \end{aligned}$$

- Hence $\triangle ABC$ is now isosceles with $AB = BC$, or $|z_1 - z_2| = |z_2 - z_3|$.

- Apply modulus to (b),

$$\begin{aligned} \frac{|z_2 - z_3|}{|z_1 - z_3|} &= \frac{|z_1 - z_3|}{|z_1 - z_2|} \\ |z_1 - z_2| |z_2 - z_3| &= |z_1 - z_3|^2 \end{aligned}$$

But $|z_1 - z_2| = |z_2 - z_3|$,

$$\begin{aligned} \therefore |z_1 - z_2|^2 &= |z_1 - z_3|^3 \\ \therefore |z_1 - z_2| &= |z_1 - z_3| \end{aligned}$$

i.e. $AC = AB$. Hence $AB = AC = BC$, and $\triangle ABC$ is equilateral.**Question 13** (Ziaziaris)

(a) i. (3 marks)

$$\begin{cases} x = a \cos \theta \\ y = b \sin \theta \end{cases}$$

Differentiating to obtain gradient of tangent:

$$\begin{aligned} \frac{dx}{d\theta} &= -a \sin \theta \\ \frac{dy}{d\theta} &= b \cos \theta \\ \frac{dy}{dx} &= \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{b \cos \theta}{-a \sin \theta} \end{aligned}$$

Hence gradient of normal is

$$m_{\perp} = \frac{a \sin \theta}{b \cos \theta}$$

Apply point-gradient formula,

$$\begin{aligned} y - b \sin \theta &= \frac{a \sin \theta}{b \cos \theta} (x - a \cos \theta) \\ by \cos \theta - b^2 \sin \theta \cos \theta &= ax \sin \theta - a^2 \sin \theta \cos \theta \\ ax \sin \theta - by \cos \theta &= a^2 \sin \theta \cos \theta - b^2 \sin \theta \cos \theta \\ \therefore ax \sin \theta - by \cos \theta &= (a^2 - b^2) \sin \theta \cos \theta \quad (\ddagger) \end{aligned}$$

ii. (2 marks)

- G occurs when normal meets x axis: substitute $y = 0$ into $(\ddagger)$:

$$\begin{aligned} ax \sin \theta - 0 &= (a^2 - b^2) \sin \theta \cos \theta \\ x &= \frac{(a^2 - b^2) \cos \theta}{a} \end{aligned}$$

- N occurs at $x = a \cos \theta$, $y = 0$ (on x axis):
- Hence $OG = \frac{(a^2 - b^2) \cos \theta}{a}$,
 $ON = a \cos \theta$:

$$\begin{aligned}\frac{OG}{ON} &= \frac{\frac{(a^2 - b^2) \cos \theta}{a}}{a \cos \theta} \\ &= \frac{a^2 - b^2}{a^2} \\ &= 1 - \frac{b^2}{a^2}\end{aligned}$$

As $b^2 = a^2(1 - e^2)$, then
 $\frac{b^2}{a^2} = 1 - e^2$:

$$\begin{aligned}\therefore \frac{OG}{ON} &= 1 - (1 - e^2) \\ &= e^2\end{aligned}$$

ii. (2 marks)

$$m_{\perp} = t^2$$

Use pt-gradient formula to find equation of normal:

$$y - \frac{c}{t} = t^2(x - ct)$$

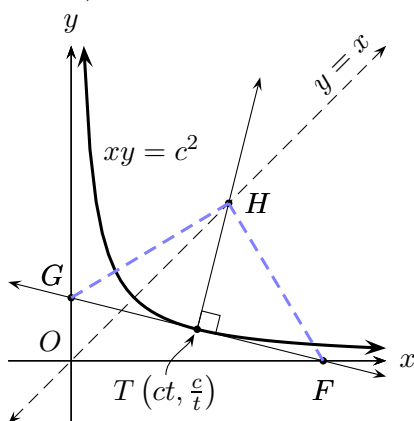
$$\underbrace{y - \frac{c}{t}}_{\times t} = \underbrace{t^2x - ct^3}_{\times t} \quad (\clubsuit)$$

$$ty - c = t^3x - ct^4$$

$$t^3x - ty = ct^4 - c$$

$$\therefore t^3x - ty = c(t^4 - 1)$$

(b) i. (3 marks)



iii. (5 marks)

- Point H : Normal meets $y = x$.
Replace y with x in $(\clubsuit)$:

$$\begin{aligned}t^3x - tx &= c(t^4 - 1) \\ x(t)(\cancel{t^2 - 1}) &= c(\cancel{t^2 - 1})(t^2 + 1) \\ \therefore x &= y = \frac{c(t^2 + 1)}{t} \quad (\heartsuit)\end{aligned}$$

- Point F : when tangent meets x axis, $y = 0$. Use $(\spadesuit)$:

$$\begin{aligned}0 - \frac{c}{t} &= -\frac{1}{t^2}(x - ct) \\ ct &= x - ct \\ x &= 2ct \\ \therefore F(2ct, 0)\end{aligned}$$

- Point G : when tangent meets y axis, $x = 0$. Use $(\spadesuit)$:

$$\begin{aligned}y - \frac{c}{t} &= -\frac{1}{t^2}(0 - ct) \\ y &= \frac{c}{t} + \frac{c}{t} = \frac{2c}{t} \\ \therefore G\left(0, \frac{2c}{t}\right)\end{aligned}$$

Use pt-gradient formula to find equation of tangent:

$$\begin{aligned}\underbrace{y - \frac{c}{t}}_{\times(-t^2)} &= \underbrace{-\frac{1}{t^2}}_{\times(-t^2)}(x - ct) \quad (\spadesuit) \\ -t^2y + ct &= x - ct \\ x + t^2y &= 2ct\end{aligned}$$

- Gradient of FH :

$$\begin{aligned}
 m_{FH} &= \frac{y_h - y_f}{x_h - x_f} = \frac{\frac{c(t^2+1)}{t} - 0}{\frac{c(t^2+1)}{t} - 2ct} \\
 &= \frac{\cancel{c}(t^2+1)}{\cancel{c}t^2 + \cancel{c} - 2ct^2} = \frac{\cancel{c}(t^2+1)}{\cancel{c}(1-t^2)} \\
 &= \frac{1+t^2}{1-t^2}
 \end{aligned}$$

- Gradient of GH :

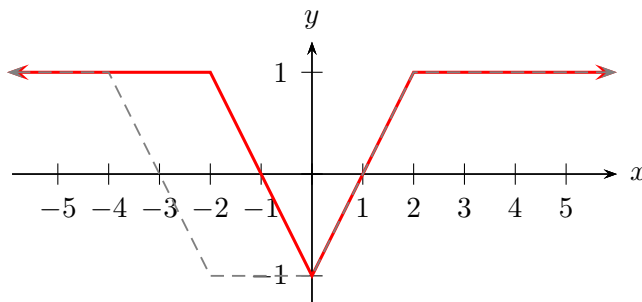
$$\begin{aligned}
 m_{GH} &= \frac{y_h - y_g}{x_h - x_g} = \frac{\frac{c(t^2+1)}{t} - \frac{2c}{t}}{\frac{c(t^2+1)}{t} - 0} \\
 &= \frac{\cancel{c}t^2 + \cancel{c} - 2c}{\cancel{c}t^2 + \cancel{c}} = \frac{\cancel{c}(t^2-1)}{\cancel{c}(1+t^2)} \\
 &= \frac{t^2-1}{1+t^2}
 \end{aligned}$$

- Multiply gradients,

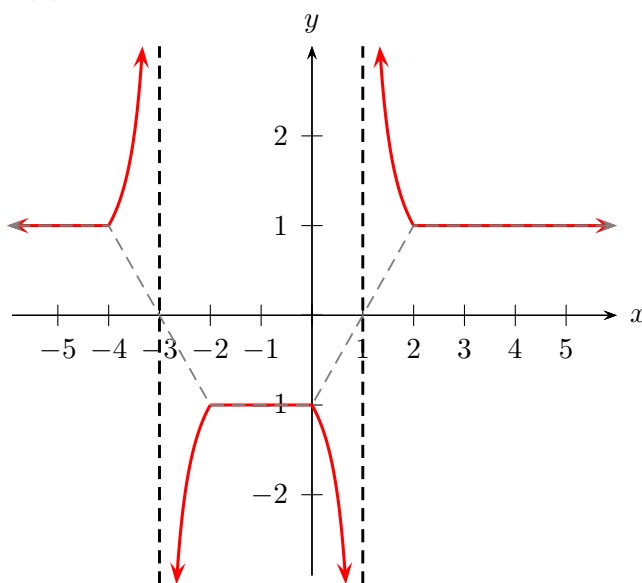
$$\begin{aligned}
 m_{GH} \times m_{GH} &= \frac{t^2-1}{\cancel{t^2+1}} \times \frac{\cancel{t^2+1}}{1-t^2} \\
 &= -1 \\
 \therefore FH &\perp GH
 \end{aligned}$$

Question 14(Lam)

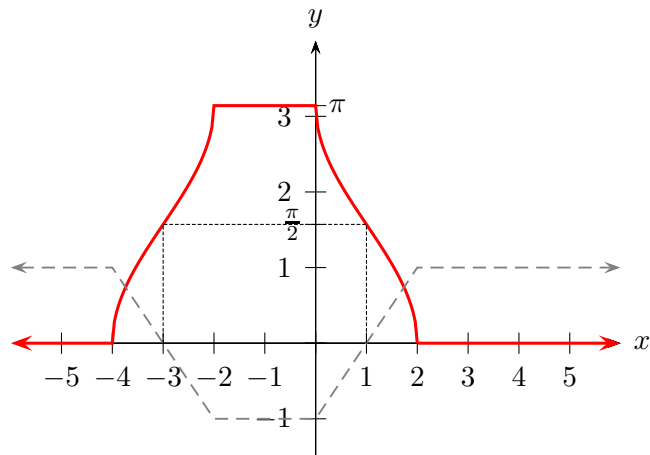
- (a) i. (2 marks) – $y = f(|x|)$. (Old curve in gray dashes)



- ii. (2 marks) – $y = \frac{1}{f(x)}$



iii. (2 marks)



(b) i. (2 marks)

$$\begin{aligned}\lceil 2.5 \rceil &= 3 \\ (2.5 + \lceil 2.5 \rceil)^2 &= (2.5 + 3)^2 \\ &= 5.5^2 \\ &= \frac{121}{4}\end{aligned}$$

ii. (3 marks)

By cases,

- From $x = -4$ to $x = -3$, $\lceil x \rceil = -3$

$$\therefore y = -3 + (x - 3)^2$$

- From $x = -3$ to $x = -2$, $\lceil x \rceil = -2$

$$\therefore y = -2 + (x - 2)^2$$

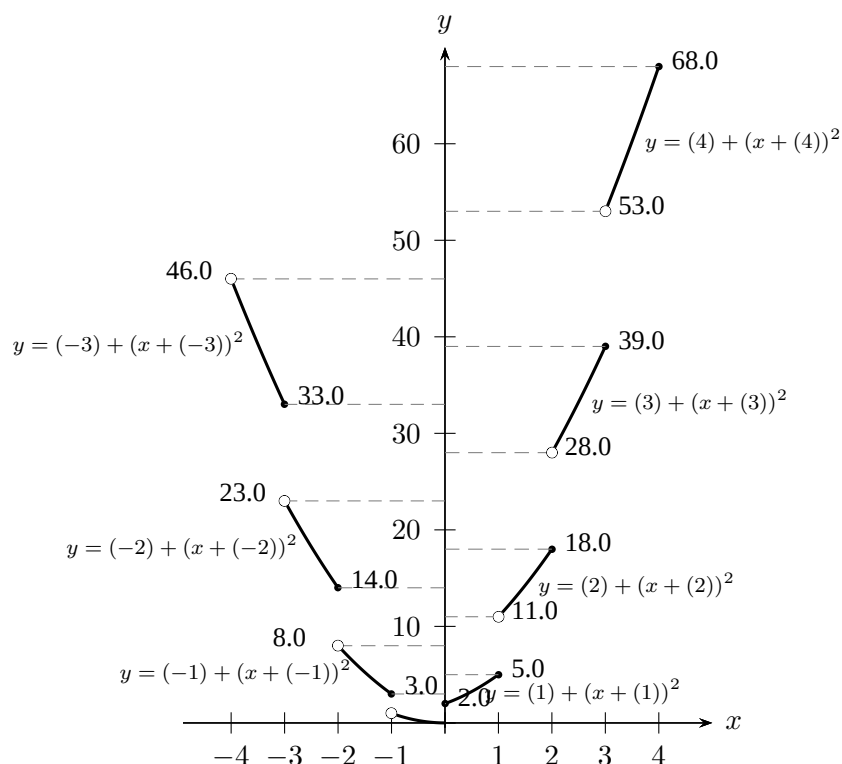
- From $x = -2$ to $x = -1$, $\lceil x \rceil = -1$

$$\therefore y = -1 + (x - 1)^2$$

- From $x = -1$ to $x = 0$, $\lceil x \rceil = 0$

$$\therefore y = 0 + (x - 0)^2$$

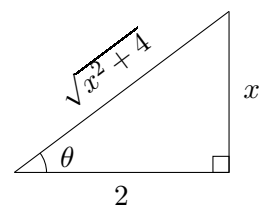
etc.



(c) (4 marks)

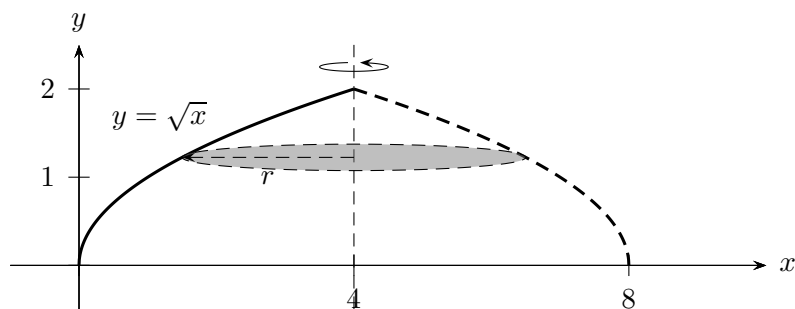
Letting $x = 2 \tan \theta$,

$$\begin{aligned}
 & \int \frac{dx}{(4 + x^2)^{\frac{3}{2}}} \\
 & \frac{dx}{d\theta} = 2 \sec^2 \theta \, d\theta \\
 & \int \frac{dx}{(4 + x^2)^{\frac{3}{2}}} = \int \frac{2 \sec^2 \theta \, d\theta}{(4 + 4 \tan^2 \theta)^{\frac{3}{2}}} \\
 & = \int \frac{2 \sec^2 \theta \, d\theta}{(4(1 + \tan^2 \theta))^{\frac{3}{2}}} \\
 & = \int \frac{2 \sec^2 \theta \, d\theta}{8 \sec^3 \theta} \\
 & = \frac{1}{4} \int \frac{1}{\sec \theta} \, d\theta \\
 & = \frac{1}{4} \int \cos \theta \, d\theta \\
 & = \frac{1}{4} \sin \theta + C \\
 & = \frac{1}{4} \frac{x}{\sqrt{x^2 + 4}} + C
 \end{aligned}$$



Question 15(Ziaziaris)

(a) i. (2 marks)



Area of disc:

$$\begin{aligned}
 r &= (4 - x) \\
 A &= \pi r^2 \\
 &= \pi (4 - x)^2
 \end{aligned}$$

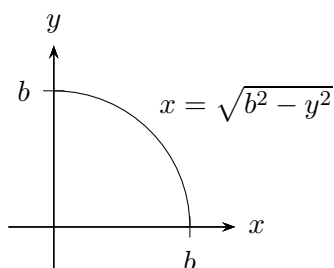
Variable sized discs run from $y = 0$ to $y = 2$, i.e. use δy for thickness

$$\begin{aligned}
 \therefore \delta V &= A \times \delta y \\
 &= \pi (4 - y^2)^2 \delta y \\
 &= \pi (16 - 8y^2 + y^4) \delta y
 \end{aligned}$$

ii. (2 marks)

$$\begin{aligned}
 V &= \pi \lim_{\delta y \rightarrow 0} \sum_{y=0}^{y=2} (16 - 8y^2 + y^4) \delta y \\
 &= \pi \int_0^2 (16 - 8y^2 + y^4) dy \\
 &= \pi \left[16y - \frac{8}{3}y^3 + \frac{1}{5}y^5 \right]_0^2 \\
 &= \pi \left(16(2) - \frac{8}{3}(8) + \frac{1}{5}(32) \right) \\
 &= \frac{256}{15} \pi
 \end{aligned}$$

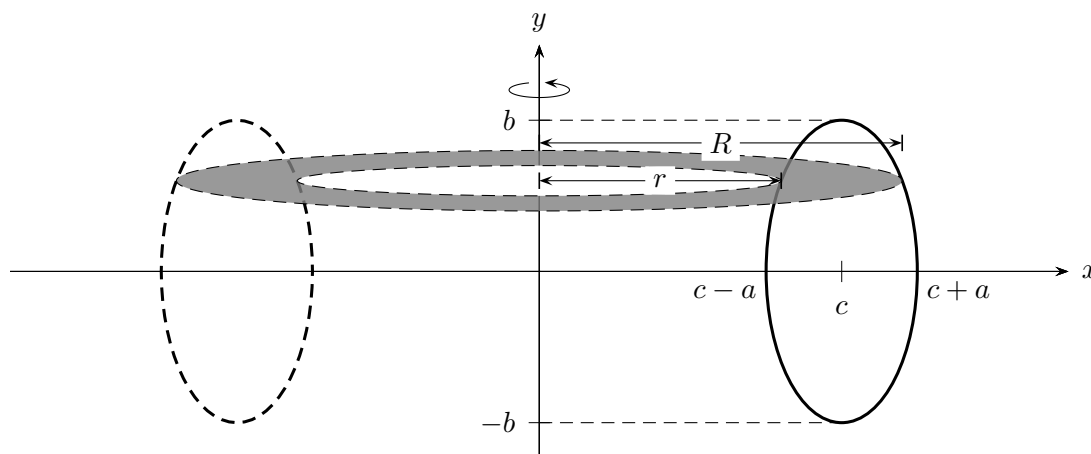
(b) i. (2 marks)



- Curve $x = \sqrt{b^2 - y^2}$ is the top half of the sideways semicircle with radius b .
- Hence $\int_0^1 \sqrt{b^2 - y^2} dy$ is the area of the quarter circle:

$$\begin{aligned}
 A &= \int_0^b \sqrt{b^2 - y^2} dy \\
 &= \frac{1}{4} \pi r^2 = \frac{1}{4} \pi b^2
 \end{aligned}$$

ii. (4 marks)



Ellipse equation:

$$\begin{aligned}\frac{(x-c)^2}{a^2} + \frac{y^2}{b^2} &= 1 \\ (x-c)^2 &= a^2 \left(1 - \frac{y^2}{b^2}\right) \\ x-c &= \pm a \sqrt{1 - \frac{y^2}{b^2}} \\ x &= c \pm a \sqrt{1 - \frac{y^2}{b^2}}\end{aligned}$$

Area of annulus:

$$\begin{aligned}A &= \pi (R^2 - r^2) \\ &= \pi (R-r)(R+r) \\ &= \pi \left(2a \sqrt{1 - \frac{y^2}{b^2}}\right) (2c) \\ &= 4\pi ac \sqrt{\frac{1}{b^2} (b^2 - y^2)} \\ &= \frac{4\pi ac}{b} \sqrt{b^2 - y^2}\end{aligned}$$

Volume element & volume generated:

- Inner radius r :

$$r = c - a \sqrt{1 - \frac{y^2}{b^2}}$$

- Outer radius R :

$$R = c + a \sqrt{1 - \frac{y^2}{b^2}}$$

$$\begin{aligned}\delta V &= A \times \delta y = \frac{4\pi ac}{b} \sqrt{b^2 - y^2} \delta y \\ V &= \frac{4\pi ac}{b} \lim_{\delta y \rightarrow 0} \sum_{y=0}^{y=b} 2 \times \sqrt{b^2 - y^2} \delta y \\ &= \frac{4\pi ac}{b} \times 2 \times \int_0^b \sqrt{b^2 - y^2} dy \\ &= \frac{4\pi ac}{b} \times 2 \times \frac{1}{2} \pi b^2 \\ &= 2\pi^2 abc\end{aligned}$$

(c) i. (2 marks)

$$\int_0^a f(x) dx$$

Let $u = a - x$, then

$$\frac{du}{dx} = -1$$

$$du = -dx$$

$$x = 0 \quad u = a$$

$$x = a \quad u = 0$$

$$\begin{aligned} \int_0^a f(x) dx &= \int_{u=a}^{u=0} f(a-u) (-du) \\ &= \int_0^a f(a-u) du \\ &= \int_0^a f(a-x) dx \end{aligned}$$

ii. (3 marks)

$$I = \int_0^{\frac{\pi}{2}} \frac{e^{\sin x}}{e^{\sin x} + e^{\cos x}} dx \left(\equiv \int_0^a f(x) dx \right) \quad (\star)$$

By using the result from above,

$$\therefore I = \int_0^{\frac{\pi}{2}} \frac{e^{\sin(\frac{\pi}{2}-x)}}{e^{\sin(\frac{\pi}{2}-x)} + e^{\cos(\frac{\pi}{2}-x)}} dx = \int_0^{\frac{\pi}{2}} \frac{e^{\cos x}}{e^{\cos x} + e^{\sin x}} dx \quad (\nabla)$$

Adding (∇) and $(\star)$,

$$\begin{aligned} 2I &= \int_0^{\frac{\pi}{2}} \frac{e^{\cos x}}{e^{\cos x} + e^{\sin x}} dx + \int_0^{\frac{\pi}{2}} \frac{e^{\sin x}}{e^{\cos x} + e^{\sin x}} dx \\ &= \int_0^{\frac{\pi}{2}} \frac{e^{\sin x} + e^{\cos x}}{e^{\sin x} + e^{\cos x}} dx \\ &= \int_0^{\frac{\pi}{2}} 1 dx \\ &= \frac{\pi}{2} \end{aligned}$$

$$\therefore 2I = \frac{\pi}{2}$$

$$I = \frac{\pi}{4}$$

Question 16(Ireland)

(a) i. (1 mark)

$$x^3 - 4x^2 + 5x + 2 = 0$$

$$\left| \begin{array}{l} (\alpha + \beta + \gamma)^2 \\ = (\alpha + \beta + \gamma)(\alpha + \beta + \gamma) \\ = \alpha^2 + \alpha\beta + \alpha\gamma \\ \quad + \beta^2 + \alpha\beta + \beta\gamma \\ \quad + \gamma^2 + \alpha\gamma + \beta\gamma \end{array} \right.$$

$$\begin{aligned} \therefore \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= \left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right) \\ &= (4^2) - 2(5) = 6 \end{aligned}$$

ii. (2 marks)

If α , β and γ are roots, then they satisfy cubic equation

$$\begin{aligned} \alpha^3 - 4\alpha^2 + 5\alpha + 2 &= 0 \\ \beta^3 - 4\beta^2 + 5\beta + 2 &= 0 \\ \gamma^3 - 4\gamma^2 + 5\gamma + 2 &= 0 \end{aligned}$$

Adding equations, and subtracting to other side,

$$\begin{aligned} \alpha^3 + \beta^3 + \gamma^3 &= 4(\alpha^2 + \beta^2 + \gamma^2) \\ &\quad - 5(\alpha + \beta + \gamma) \\ &\quad - 6 \\ &= 4(6) - 5(4) - 6 \\ &= -2 \end{aligned}$$

(b) i. (3 marks)

$$P(x) = (x^2 - a^2)Q(x) + (px + q)$$

Evaluating at $x = a$,

$$P(a) = 0 + pa + q = ap + q \quad (\blacksquare)$$

Evaluating at $x = -a$,

$$P(-a) = 0 - pa + q = -ap + q \quad (\blacktriangle)$$

Adding $(\blacksquare)$ and $(\blacktriangle)$,

$$\begin{aligned} P(a) + P(-a) &= 2q \\ \therefore q &= \frac{1}{2}[P(a) + P(-a)] \end{aligned}$$

Subtracting $(\blacksquare)$ and $(\blacktriangle)$,

$$\begin{aligned} P(a) - P(-a) &= 2ap \\ \therefore p &= \frac{1}{2a}[P(a) - P(-a)] \end{aligned}$$

ii. (2 marks)

$$\begin{aligned} P(x) &= x^n - a^n \\ &= (x^2 - a^2)Q(x) + (px + q) \end{aligned}$$

Finding p :

$$\begin{aligned} P(a) &= a^n - a^n = 0 \\ P(-a) &= (-a)^n - a^n \\ &= (-1)^n a^n - a^n \\ p &= \frac{1}{2a}[P(a) - P(-a)] \\ &= \frac{1}{2a}[0 - ((-1)^n a^n - a^n)] \\ &= \frac{1}{2a}[a^n - (-1)^n a^n] \end{aligned}$$

When n is even, $(-1)^n = 1$,

$$p = 0$$

When n is odd, $(-1)^n = -1$,

$$\begin{aligned} p &= \frac{1}{2a}[a^n - (-1)a^n] \\ &= \frac{1}{2a}(+2a^n) \\ &= a^{n-1} \end{aligned}$$

Finding q ,

$$\begin{aligned} q &= \frac{1}{2}[P(a) + P(-a)] \\ &= \frac{1}{2}[0 + ((-1)^n a^n - a^n)] \\ &= \frac{1}{2}[(-1)^n a^n - a^n] \end{aligned}$$

When n is even, $(-1)^n = 1$,

$$q = \frac{1}{2} \times 0 = 0$$

When n is odd, $(-1)^n = -1$,

$$q = \frac{1}{2} [-2a^n] = -a^n$$

Hence when n is even, the remainder is zero, whilst when n is odd, the remainder is

$$R(x) = a^{n-1}x - a^n = a^{n-1}(x - a)$$

(c) i. (2 marks)

$$\begin{cases} y = \frac{c^2}{x} & (1) \\ (x-1)^2 + y^2 = 1 & (2) \end{cases}$$

Solve simultaneously by substituting (1) to (2):

$$\begin{aligned} (x-1)^2 + \left(\frac{c^2}{x}\right)^2 &= 1 \\ \times x^2 & \quad \times x^2 \\ x^2(x-1)^2 + c^4 &= x^2 \end{aligned}$$

Since the curves *touch* at Q , the x coordinate of (β) of Q is a repeated real root of the equation. As there are no further intersections, the equation has no other real roots. Hence the remaining two roots are non-real complex conjugate roots, as the equation has real coefficients.

ii. (3 marks)

- If $x = \beta$ is a double root of $P(x) = 0$, then $x = \beta$ is also a root of $P'(x) = 0$:

$$\begin{aligned} P(x) &= x^2(x-1)^2 - x^2 + c^4 = 0 \\ &= x^2(x^2 - 2x + 1) - x^2 + c^4 \\ &= x^4 - 2x^3 + c^4 \end{aligned}$$

$$\begin{aligned} P'(x) &= 4x^3 - 6x^2 \\ &= 2x^2(2x - 3) \end{aligned}$$

As $P(\beta) = 0$, then $P'(\beta) = 0$ and $\beta \neq 0$

$$\begin{aligned} \therefore 2x - 3 &= 0 \\ x = \beta &= \frac{3}{2} \end{aligned}$$

Finding c^2 : substitute $x = \frac{3}{2}$ into quartic

$$\begin{aligned} \left(\frac{3}{2}\right)^4 - 2\left(\frac{3}{2}\right)^3 + c^4 &= 0 \\ c^4 &= \frac{27}{16} \\ \therefore c^2 &= \frac{3\sqrt{3}}{4} \end{aligned}$$

iii. (2 marks)

Equation of common tangent: find $\frac{dy}{dx}$.

$$\begin{aligned} y &= \frac{c^2}{x} = c^2 x^{-1} \\ \frac{dy}{dx} &= -c^2 x^{-2} \Big|_{x=\frac{3}{2}} \\ & \quad c^2 = \frac{3\sqrt{3}}{4} \\ &= -\frac{3\sqrt{3}}{4} \times \frac{4}{9} \\ &= -\frac{\sqrt{3}}{3} = -\frac{1}{\sqrt{3}} \end{aligned}$$

At the point of contact, $x = \frac{3}{2}$:

$$\begin{aligned} y^2 &= 1 - (x-1)^2 \\ y &= \sqrt{1 - \left(\frac{3}{2} - 1\right)^2} = \frac{\sqrt{3}}{2} \end{aligned}$$

Apply point-gradient formula,

$$\begin{aligned} y - \frac{\sqrt{3}}{2} &= -\frac{1}{\sqrt{3}} \left(x - \frac{3}{2}\right) \\ y &= -\frac{1}{\sqrt{3}}x + \frac{\sqrt{3}}{2\sqrt{3}} + \frac{\sqrt{3}}{2} \\ & \quad \nearrow^{\sqrt{3}} \\ y &= -\frac{1}{\sqrt{3}}x + \sqrt{3} \end{aligned}$$